

OPTION PANDA — Scoring System & Strategy Guide

A complete walkthrough of how the dashboard rates every ticker, what each rule measures, and how the strategy presets combine them.

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Executive Summary

OPTION PANDA scores every ticker in its ~2,500-symbol universe against a library of 25+ technical, fundamental, options-flow, and sector rules. Each rule contributes a fixed point weight when its condition fires. The total is clamped to a 0–164 band; tickers at or above the user-tuned Min Score threshold are marked GO.

Five built-in strategy presets layer 'must-pass' filters on top of the score so the user can shift the scanner from broad discovery (any GO) to evidence-gated rarity (HIGH CONVICTION). Each preset is also paired with its own minimum score, calibrated against the observed distribution (peaks 140–160 on a ~164-pt practical max as of the 2026-04-30 retune).

The scoring weights are NOT arbitrary. Every rule was derived from a named academic source, an industry-standard methodology, or a council-reviewed backtest. The most heavily-weighted rules (JT 12-1, Minervini Trend Template, Pocket Pivot, U/D Volume Ratio) are the ones with the strongest replicated edge in the literature.

This document covers: the full rule catalog (math + source + weight), the GO-decision logic and ranking tiebreakers, every strategy preset with its filter set and intent, calibration history, and known limitations.

1. The Scoring Engine

Every ticker enters `scoreIt()` and runs through ~25 rules grouped into five sections plus overlays. Each rule has a fixed point weight, fires independently, and never short-circuits — a single failing rule doesn't disqualify a ticker, only reduces its total score.

Practical max score = 164 points. Empirical peak in normal markets = 140–160. Default GO threshold = 140 (~85% of practical max).

2. Full Rule Catalog

Every rule the dashboard ships. Weight, what it tests, and the math / academic source. Rules grouped by section.

Section A — Technical Momentum (Tradier bulk quote)

Rule	Wt	Tests	Math / Source
RSI 40-70	6	Permissive momentum band — not oversold, not extreme.	$RSI(14) \geq 40$ and ≤ 70 .
RSI 50-65 (Cardwell)	6	Cardwell range-shift zone — early entry inside a bull regime.	$RSI(14) \geq 50$ and ≤ 65 . Bonus on top of RSI 40-70.
MACD Hist +	6	MACD histogram > 0 — momentum is positive.	Standard 12/26/9 MACD. $macdHist > 0$.
ADX > 25	6	Confirmed trend strength (Wilder's threshold).	$ADX(14) > 25$.
ADX 15-25 + $DI+ > DI-$	6	Emerging-trend Wilder original signal — $DI+$ leading $DI-$ with ADX rising.	$ADX(14) \in [15, 25]$ and $DI+ > DI-$. Fires on detail-load only.
EMA20 $>$ EMA50	10	Short-term trend above medium-term trend.	$EMA(20) > EMA(50)$ on daily closes.

Rule	Wt	Tests	Math / Source
TF Aligned 2/3	12	At least 2 of 3 timeframes (daily / hourly / 15-min) point the same direction.	Tradier multi-timeframe quote aggregation. Count of aligned tf ≥ 2 .
Sector PF > 2	5	Ticker's static sector profit-factor > 2 in long-call backtest.	SECTOR_PF lookup table from prior backtests.
52Wk Hi Prox <15%	8	Price within 15% of the 52-week high — universe-pass for breakout setups.	$\text{price} \geq \text{week52hi} \times 0.85$.
52Wk Hi Prox <5% (G-H)	6	George-Hwang JF 2004 effect: returns concentrate within 5% of 52-wk hi.	$\text{price} \geq \text{week52hi} \times 0.95$. Bonus on top of 15% rule.
RS > SPY (21d)	10	21-day relative strength positive vs SPY. Falls back to 5-day if 21d missing.	Total return(21d, ticker) – Total return(21d, SPY) > 0. 5d fallback only when 21d unavailable.
Price > EMA200	6	Above the long-term moving average — long-trend bias.	$\text{price} > \text{EMA}(200)$. Falls back to $\text{EMA}20 > \text{EMA}50$ if EMA200 missing.

Section B — Fundamentals / FMP

Rule	Wt	Tests	Math / Source
Analyst Revisions $\uparrow$	12	≥ 2 analysts have raised estimates and there are recent upgrades.	FMP analyst estimates: $\text{revisionsUp} \geq 2$ AND $\text{upgradesLast30} > 0$. Long-calls bias: upward revisions outperform downward revisions (asymmetry — weighted higher than original +12 here, see commit log).
Analyst PT Exists	8	A consensus analyst price target is published.	FMP price target: ptConsensus present. Light fundamental confirmation.

Section C — GEX / Options Flow

Rule	Wt	Tests	Math / Source
GEX Call Flow (55–69%)	4	Emerging call-side OI bias in the 30-DTE chain (constructive but not crowded).	GEX call% of OI $\in [55, 69]$.
GEX Call Flow (70–74%)	6	Strong but not overcrowded call-side flow.	GEX call% of OI $\in [70, 74]$.
GEX Call Flow ($\geq 75\%$)	-6	Overcrowded call-side OI — contrarian signal, dealers pin/fade.	GEX call% of OI ≥ 75 . PENALTY (council 3/3 finding: extreme positioning is reliably negative on a 1-week horizon).

Section D — Momentum Acceleration (history-based)

Rule	Wt	Tests	Math / Source
MFI(14) > 50	10	Money Flow Index above midline — rising volume-weighted momentum.	$\text{MFI}(14) > 50$ (RSI mid-line analog with volume). Council 2026-04-29: 14-period is Chande standard; 7 was too noisy.

Rule	Wt	Tests	Math / Source
CMO(9) > 50	7	Chande Momentum Oscillator > 50 (Chande's original 1993 threshold).	CMO(9) > 50. 9-period is Chande-documented swing horizon.

Section E — Evidence-Based Swing Rules (history-based)

Rule	Wt	Tests	Math / Source
Minervini Trend Template	15	Stage 2 trend confirmation — 6 of 8 SEPA criteria computable from daily bars.	All six must pass: price > MA50; price > MA150; price > MA200; MA50 > MA150 > MA200; MA200 rising vs 22 bars ago; ≥30% above 52-wk low. Source: Mark Minervini 'Trade Like a Stock Market Wizard' (2013) SEPA.
Pocket Pivot	15	O'Neil / Morales-Kacher institutional accumulation signal.	Today's volume > largest down-day volume of prior 10 days AND close > MA10 AND close in upper half of day's range AND ≤5% extended from MA10.
U/D Volume Ratio (50d)	+12 / +8 / +4	Up-volume vs down-volume ratio over trailing 50 sessions (O'Neil IBD).	$\Sigma(\text{volume on up-close days}) / \Sigma(\text{volume on down-close days})$ over 50d. $\geq 2.0 = +12$ (strong sponsorship); $\geq 1.5 = +8$ (accumulation); $\geq 1.25 = +4$ (mild); $< 1.25 = 0$.
VCP — ATR Contraction	12	Volatility Contraction Pattern: supply drying up before breakout.	ATR(14) today / ATR(14) 30 sessions ago < 0.70. Minervini methodology.
BBW Squeeze (<p20 of 120d)	10	Bollinger Band-width compressed in the bottom 20% of its trailing range.	BB(20, 2 σ) width / mean. Today's value ≤ 20th-percentile of last 120 days.
20d Donchian Breakout + Vol	12	Turtle Traders breakout — Dennis / Eckhardt 1983-88 methodology.	today close > 20-day high AND today volume ≥ 1.5 × 50-day avg volume.
JT 12-1 Momentum	20	Cross-sectional momentum, skip last month (Jegadeesh-Titman 1993).	Return from t-252 to t-21 (12 months, skipping recent month). Threshold: ≥ +25% (top-quintile proxy). Single strongest replicated factor in equity literature.

Overlays & Penalties

Rule	Wt	Tests	Math / Source
PROVEN ticker	8	Ticker is on the curated PROVEN good-mover list.	Hand-maintained list. +8 flat bonus.
AVOID ticker	-15	Ticker is on the curated AVOID list (chronic chop / scams / fundamentals).	Hand-maintained list. -15 flat penalty and forced NO-GO.
Strong Sector	+8 / -8	Daily + intraday theme RS overlay vs SPY.	Combined daily + intraday theme score. $\geq 75 = +8$ (top quartile); $\leq 25 = -8$ (bottom quartile). Intraday rotation can trigger same-day.
Broken Trend (penalty)	-5	Price below EMA50 — medium-term trend broken; calls fight gravity.	price < EMA(50). Penalty only — no positive credit (already implicit in EMA20>EMA50 and Price>EMA200 rules).

3. GO Decision & Ranking

How the GO flag is set

After all rules fire, the raw score is clamped to the 0–164 band. Penalties (Broken Trend –5, weak sector –8, AVOID –15) can drive the subtotal negative; the clamp forces it to 0 so the threshold comparator doesn't see nonsensical scores.

GO threshold is read from the Min Score input on the scanner tab. Default 140 was calibrated 2026-04-30 against the observed distribution (peaks 140–160 on a ~164-pt practical max). Lowering the threshold widens the GO list; raising it tightens. Strategy presets override the default with their own per-strategy min-score (see preset table).

AVOID-list tickers can NEVER be GO regardless of score — the AVOID flag forces NO-GO downstream of the score clamp.

Tiebreaker order

Priority	Field
1. Score	Primary sort — descending.
2. techScore	Sum of all Section A + D + E rules. Stronger technical setup wins when total scores tie.
3. relStr5d	5-day relative-strength % vs SPY. Stronger short-term momentum wins.
4. Theme dailyScore	Sector / theme strength from theme_scores.json. Stronger sector context wins.
5. Volume	Daily volume — liquidity floor as the final tie-break.

4. Strategy Presets

Five preset combinations layer must-pass filters on top of the score, plus their own minimum score threshold. Click any preset chip in the STRATEGY row to apply.

★ HIGH CONVICTION — Min Score 140

Intent. Rarest, highest-quality setups. Stacks the two highest-IC scoring rules (JT 12-1 at +20 and Minervini at +15) as MUST-PASS instead of merely contributing to the score. A ticker can score 160+ on other rules but still fail this preset if either trend confirmation is missing.

Must-pass filters.

- PROVEN ticker
- Analyst Revisions ↑
- GEX Call Heavy ($\geq 55\%$)
- JT 12-1 Momentum ($\geq 25\%$, top-quintile proxy)
- Minervini Trend Template (6/6)

Expected matches. 0–5 matches per scan in normal market conditions.

Efficacy / rationale. Designed for swing entries where the user wants maximum signal confluence before paying premium. Backed by: JT 12-1 = strongest replicated cross-sectional factor (Asness-Moskowitz-Pedersen 2013); Minervini SEPA = Stage 2 trend filter (Minervini 2013); analyst revisions = upward earnings surprise lead indicator (Stickel 1991, Bradshaw 2004); PROVEN ticker bias = curated list of names with favorable historical long-call distribution.

▲ BREAKOUT — Min Score 125

Intent. Chart-pattern-driven breakout buying with momentum + leadership. Donchian and VCP are the two highest-IC pure-pattern breakout rules; requiring BOTH filters out fake-out names that just look extended without the underlying volatility-contraction or volume-confirmation signal.

Must-pass filters.

- 52Wk Hi Prox $< 15\%$
- RS > SPY
- MACD Hist +
- TF Aligned 2/3
- 20d Donchian Breakout + Vol
- VCP — ATR Contraction

Expected matches. 5–25 matches per scan, market-regime dependent.

Efficacy / rationale. Two-mode confirmation: (i) chart-level breakout via 20-day Donchian high with 1.5× volume (Turtle Traders methodology, Dennis-Eckhardt 1983-88); (ii) volatility contraction prior to the break (Minervini VCP — ATR(14) ratio < 0.70). Combined with relative strength + MACD positive + multi-timeframe alignment to screen out reversion-to-mean candidates.

■ MOMENTUM — Min Score 120

Intent. Confirmed trending stocks with sector tailwind, gated by the academic gold-standard 12-1 cross-sectional momentum signal. Making JT 12-1 MUST-PASS turns this preset from 'trendy' into 'trendy AND riding the strongest published factor'.

Must-pass filters.

- MACD Hist +
- Price > EMA200
- RS > SPY
- Strong Sector
- JT 12-1 Momentum

Expected matches. 5–30 matches per scan during trending markets, fewer in chop.

Efficacy / rationale. Primary edge is the 12-1 cross-sectional momentum (Jegadeesh-Titman 1993, Asness-Moskowitz-Pedersen 2013). Layered with sector rotation (theme overlay top quartile) and standard trend confirmation (MACD, EMA200, RS). Lower min-score (120) reflects looser breakout requirement vs. BREAKOUT preset — more inclusive of mid-trend continuations.

◆ BEST ITM CALLS — Min Score 145

Intent. Deep-ITM (δ 0.70–0.90) calls on names that pass ALL THREE of the highest-IC scoring rules. You're paying real premium for deep-ITM contracts — only setups confirmed by 12-month momentum + Stage 2 trend template + institutional accumulation should graduate.

Must-pass filters.

- MACD Hist +
- Price > EMA200
- RS > SPY
- JT 12-1 Momentum
- Minervini Trend Template
- Pocket Pivot
- Price $\geq$ \$5
- Avg Volume $\geq$ 1,000,000
- Vol ratio $\geq$ 100% (today vs avg)
- Max option price $\leq$ \$8

Expected matches. 0–8 matches per scan; very selective.

Efficacy / rationale. Highest-signal preset by construction. Designed to surface names where (a) you can afford the contract ($\leq$ \$8 ceiling), (b) liquidity is real ($\geq$ \$5 underlying, $\geq$ 1M avg volume, today's volume $\geq$ average), and (c) the trend / momentum / accumulation signals all agree. Pocket Pivot is the institutional-buying flag — distinguishes true accumulation from just-rallying-on-air.

▲ PRE/POST MARKET — Min Score 140

Intent. Extended-hours gappers screened with the SAME filters as HIGH CONVICTION, then re-sorted by absolute gap %. Use any time the extended-hours session is active or on weekends.

Must-pass filters.

- PROVEN ticker
- Analyst Revisions ↑
- GEX Call Heavy (≥55%)
- JT 12-1 Momentum
- Minervini Trend Template
- Sort: |gap %| descending (overrides standard score sort)

Expected matches. 0–5 matches; same selectivity as HIGH CONVICTION, re-ordered by gap size.

Efficacy / rationale. Behaviorally important: the user explicitly wanted to filter out random gappers and ONLY surface big movers that also pass the high-conviction gate. Tradier's `change\_percentage` reflects extended-hours quotes during 3–8:30am CST (pre) and 3–7pm CST (post); on weekends it shows the most recent extended-hours print.

Strategy comparison at a glance

Preset	Min Score	# Must-Pass	Expected hits
★ HIGH CONVICTION	140	5	0–5 matches per scan in normal market conditions.
▲ BREAKOUT	125	6	5–25 matches per scan, market-regime dependent.
■ MOMENTUM	120	5	5–30 matches per scan during trending markets, fewer in chop.
◆ BEST ITM CALLS	145	10	0–8 matches per scan; very selective.
▲ PRE/POST MARKET	140	5	0–5 matches; same selectivity as HIGH CONVICTION, re-ordered by gap size.

5. Calibration History

The rule set and thresholds were derived through a structured review process ("council" — domain advisors for quant, microstructure, risk, behavioral, engineering). Each entry below is a dated change to the scoring system with the rationale.

2026-04-29 council overhaul (drops)

Removed three anti-signal rules: RSI(7) Accel ($\approx 70\%$ noise per Connors), 5d %chg Accel (Lehmann 1990 short-term reversal), Vol > Avg flat threshold (too crude — replaced by U/D Volume Ratio).

2026-04-29 council overhaul (retunes)

Tightened existing rules to evidence-based parameters: RSI band $\rightarrow$ Cardwell range-shift; MFI(7) $\rightarrow$ MFI(14) > 50; CMO zero-cross $\rightarrow$ CMO(9) > 50; 5d RS $\rightarrow$ 21d RS; 52-wk Hi <15% $\rightarrow$ <15% pass + <5% George-Hwang bonus.

2026-04-29 council overhaul (additions)

Added 7 evidence-based rules: Minervini Template (+15), Pocket Pivot (+15), U/D Volume Ratio (+4/+8/+12), VCP (+12), BBW Squeeze (+10), Donchian (+12), JT 12-1 Momentum (+20). Practical max raised from ~ 110 to ~ 164 .

2026-04-30 GO threshold retune

Default GO threshold raised 100 $\rightarrow$ 150, then re-calibrated 2026-05-03 to 140 against the observed distribution. New rules rarely hit max simultaneously — distribution peak sits 140–160, not 165+.

2026-05-01 GEX retune

GEX Call Flow rule moved from flat +6 to tiered with extreme-positioning PENALTY: 55–69% = +4 (emerging), 70–74% = +6 (strong), $\geq 75\%$ = –6 (overcrowded contrarian). Backed by council finding that $\geq 75\%$ OI on calls reliably mean-reverts on a 1-week horizon.

2026-05-04 Analyst Revisions bump

Analyst Revisions $\uparrow$ bumped +12 $\rightarrow$ +16 for the long-calls-only user. Upward revisions outperform downward revisions asymmetrically — call-only strategies should weight bullish signals more heavily.

2026-05-03 Strong Sector bump

Strong Sector $\pm 5 \rightarrow \pm 8$. Bullish unhedged bets depend more heavily on sector tailwind than hedged or directional-pair strategies do.

6. Efficacy & Known Limitations

Honest framing: the rule weights and thresholds are expert-judgment-informed-by-literature, not empirically backed by a large-sample backtest of THIS COMBINED SYSTEM. Treat the score as a structured checklist, not a guaranteed edge.

- Single-leg long-call bias. The scoring is explicitly tuned for long-call swing trades. It is not validated for short premium, spreads, puts, or longer-than-50-DTE positions.
- Backtest $n=19$ trades (2026-04-29 council review). The empirical edge of the combined system is small and not statistically separated from noise. Two-thirds of cumulative P&L came from one outlier (POET 12.5C, 8-DTE, +\$1,225). Treat all rule weights as expert-judgment-with-priors, not empirically-validated weights.

- Some rules require ≥ 210 daily bars (Minervini, Donchian, U/D Volume, BBW). Fresh IPOs or newly-uplisted symbols are flagged with `dataGap = true` and treated as 'unknown', NOT as failed. Strategies that require those rules MUST-PASS will simply exclude such names.
- GEX data is sourced from a separate options-flow tab (runs once per scan + scheduled at 9:35 / 12:00 / 15:00 ET on weekdays). If GEX hasn't run yet the GEX Call Flow rule scores 0 — neither bonus nor penalty. Strategies that MUST-PASS GEX will exclude all tickers until GEX data lands.
- Cross-sectional ranking (JT 12-1's true form) is approximated via a hard +25% threshold rather than ranking against the universe simultaneously. This is a simplification — true cross-sectional momentum would require computing returns for every scanned ticker before classifying. Threshold is set near the historical top-quintile cut.
- Theme / sector overlay depends on `theme_scores.json` which is regenerated by a nightly GitHub Actions cron. If the cron fails (network blip, Tradier outage, etc.), the Strong Sector rule pass-throughs (no bonus / no penalty) so the rest of the scoring still works.

7. End-User Playbook

Step-by-step for someone opening the dashboard for the first time.

1. Set your API keys

Open the API tab and enter your Tradier proxy URL + live-mode token, and your FMP API key. Without an FMP key, Section B rules score 0 — the rest still works.

2. Pick a strategy preset

Tap STRATEGY at the top of the scanner. Start with HIGH CONVICTION to see the most selective gate, or BREAKOUT / MOMENTUM if you want more candidates. CLEAR returns to the default Min Score 140 with no must-pass filters.

3. Run the scan

Tap ■ SCAN. The bulk pass scores all ~2,500 tickers in 30–60s. The foreground enrichment pass adds Tradier history + FMP data to the top 60. A background pass extends Tradier history to ranks 61–200 over the next ~5 minutes; you'll see scores update live.

4. Read the results table

Sortable columns: Score, Tech, Fund, Price, Change %, Theme, Sector, GEX flow, etc. Click any row to open the detail pane with the full rule-by-rule breakdown — every passing rule with its point value, every failing rule with what it would have scored.

5. Use Export Matrix or Export GOs

EXPORT GOs copies a comma-separated list of every ticker that's GO under the current filter. EXPORT MATRIX in the Portfolio tab pulls all positions into a Google Sheet for sharing.

6. Tune Min Score if needed

If a preset returns 0 matches in a quiet market, drop Min Score by 10–20 to widen the GO band. If it returns 50+ in a hot market, raise it to tighten. The score distribution shifts with regime.

Appendix — Glossary

Term	Definition
GO / NO-GO	Binary classification output by scoreIt. GO = score $\geq$ Min Score AND not AVOID-listed.
Min Score	User-tunable input on the scanner tab (default 140). Strategy presets override with their own per-strategy minimum.
Must-Pass	Filter applied AFTER scoring. A ticker that scores 200 but fails a must-pass filter (e.g. no Analyst Revisions $\uparrow$) is excluded from the preset's GO list.
techScore	Sum of all Section A + D + E rule weights that fired for this ticker. Used as the second tiebreaker.
fundScore	Sum of Section B + Strong Sector. Surfaces analyst + sector tailwind contribution.
GEX	Gamma exposure — dealer-hedging pressure derived from the 30-DTE options chain. Call Heavy = $\geq 55\%$ of OI on the call side.
JT 12-1	Jegadeesh-Titman 12-1 momentum — return from t-252 to t-21 (skip last month).
VCP	Volatility Contraction Pattern — Minervini's ATR-shrinkage precursor to breakout.
Donchian	Turtle Traders breakout — close above prior 20-day high on 1.5 $\times$ + volume.
BBW	Bollinger Bandwidth — (upper – lower) / mean. Compressed when in the bottom 20% of the trailing 120-day range.
PROVEN ticker	Hand-curated list of names with historically favorable long-call P&L. +8 bonus on score.
AVOID ticker	Hand-curated AVOID list. -15 penalty + forced NO-GO.
Theme overlay	Sector/theme RS from theme_scores.json (nightly cron). Drives the Strong Sector ± 8 rule.
Intraday RS	Live theme RS computed from member changePct vs SPY during the day so rotation triggers same-day.

Source of truth: scoreIt() and STRATEGY_MODES in index.html. When those change, update build_scoring_report.py and rebuild.